AI-Powered Multi-Agent Energy Research System

A Project-II Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

BY

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Anuroop Patidar - EN22CS301176
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Under the Guidance of

Prof. Akshay Saxena
Prof. Hemlata Patel
Prof. Ajeet Singh Rajput

MEDICAPS UNIVERSITY

Department of Computer Science & Engineering
Faculty of Engineering
MEDICAPS UNIVERSITY, INDORE- 453331

JAN-JUNE 2026

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Report Approval

The project work **"AI-Powered Multi-Agent Energy Research System"** is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approved any statement made, opinion expressed, or conclusion drawn there in; but approve the "Project Report" only for the purpose for which it has been submitted.

Internal Examiner

Name:

Designation

Affiliation

External Examiner

Name:

Designation

Affiliation

Declaration

I/We hereby declare that the project entitled **"AI-Powered Multi-Agent Energy Research System"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and engineering completed under the supervision of **Prof. Akshay Saxena, Prof. Hemlata Patel, Prof. Ajeet Singh Rajput,** Faculty of Engineering, Medicaps University Indore is an authentic work.

Further, I/we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Signature and name of the student(s) with date

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Certificate

We, **Prof. Akshay Saxena, Prof. Hemlata Patel, Prof. Ajeet Singh Rajput** certify that the project entitled "**AI-Powered Multi-Agent Energy Research System"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications by **Ansh Upadhyay, Anuroop Patidar, Arnika Jain, Arnoday Soni, Aryan Singh** is the record carried out by him/them under my/our guidance and that the work has not formed the basis of award of any other degree elsewhere.

Prof. Akshay Saxena	<Name of External Guide (If any)>
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Abstract

The rapid growth of digital information has made manual research increasingly time-consuming and inefficient. This project presents the design and implementation of **an AI-powered Energy Research Application** that automates the process of web-based research, analysis, and report generation.

The system leverages **Generative Artificial Intelligence (GenAI)** and **Agentic AI** architecture, where multiple specialized agents collaborate to perform tasks such as web searching, content scraping, data analysis, and report summarization. The application integrates modern technologies including React (Vite) for frontend development, Node.js (Express) as an API gateway, and FastAPI (Python) for backend processing. Advanced AI models such as HuggingFace Transformers (Flan-T5) and Sentence Transformers are used for natural language processing and semantic search.

The system also incorporates **FAISS** (Facebook AI Similarity Search) for efficient vector-based retrieval and uses **JSON-based storage** for maintaining research history. To ensure scalability and real-world applicability, the application is containerized using **Docker** and deployed on **AWS EC2** with a fully automated **CI/CD** pipeline using **GitHub Actions**.

The developed system successfully automates the research workflow, reduces manual effort, and generates structured reports from user queries. The project demonstrates the integration of AI, web technologies, and DevOps practices into a unified, scalable, and production-ready system.

Keywords:

Artificial Intelligence, Generative AI, Multi-Agent Systems, Web Scraping, Natural Language Processing, Semantic Search, FAISS, Docker, CI/CD, AWS EC2

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
GenAI	Generative Artificial Intelligence
LLM	Large Language Model
API	Application Programming Interface
UI	User Interface
UX	User Experience
CI/CD	Continuous Integration / Continuous Deployment
EC2	Elastic Compute Cloud
FAISS	Facebook AI Similarity Search
NLP	Natural Language Processing
HTTP	HyperText Transfer Protocol
JSON	JavaScript Object Notation

Chapter 1: Introduction

1.1 Introduction

In the modern digital era, the volume of information available on the internet has grown exponentially. While this abundance of data provides immense opportunities for knowledge discovery, it also introduces challenges in efficiently extracting, analyzing, and summarizing relevant information. Traditional research methods require significant manual effort, time, and domain expertise to gather and process information from multiple sources.

To address these challenges, the **Energy Research App** is developed as an intelligent, AI-powered system that automates the process of web-based research and report generation. The application leverages **Generative Artificial Intelligence (GenAI)**, **Agentic AI (multi-agent systems)**, and modern **web technologies** to simulate the workflow of a human research analyst. Additionally, the system is deployed using **Docker containers**, automated through **CI/CD pipelines using GitHub Actions**, and hosted on **AWS EC2**, demonstrating real-world DevOps practices.

1.2 Literature Review

More recently, AI-based tools such as conversational agents and research assistants have emerged. These systems utilize Large Language Models (LLMs) to generate human-like responses and summaries. However, most existing solutions exhibit certain limitations:

- Lack of end-to-end automation of the research pipeline
- Limited ability to scrape and analyze real-time web data
- Absence of multi-agent collaboration mechanisms
- No persistent storage and semantic retrieval of research outputs

Research in Agentic AI systems has introduced the concept of dividing complex tasks into smaller subtasks handled by specialized agents.

1.3 Objectives

The primary objectives of this project are as follows:

- To implement a **multi-agent AI architecture** for task distribution
- To integrate **Generative AI models** for report generation
- To develop a user-friendly interface for interaction and visualization
- To deploy the system using **Docker and AWS cloud infrastructure**
- To implement a **CI/CD pipeline** for continuous integration and deployment

1.4 Significance

The key contributions include:

- **Automation of Research Process:** Reduces manual effort and time required for gathering and analyzing information

- **Integration of AI Technologies:** Demonstrates practical use of Generative AI and Agentic AI
- **Scalable Architecture:** Designed using modular components for easy scalability and maintenance
- **Real-World Deployment:** Implements industry-standard DevOps practices such as containerization and CI/CD

1.5 Research Design

The system follows a modular and layered design approach, ensuring separation of concerns.

Design Approach:

- **Frontend Layer:** Handles user interaction and displays results
- **API Layer:** Acts as a communication bridge between frontend and backend
- **Backend Layer:** Processes requests and manages AI operations
- **Agent Layer:** Executes specialized tasks such as research, analysis, and summarization
- **Storage Layer:** Stores reports and embeddings

1.6 Source of Data

The data used in this system is collected dynamically from online sources. The primary sources include:

- **Search Engines:** DuckDuckGo for retrieving relevant links
- **Web Pages:** Scraped using BeautifulSoup and HTTP requests
- **Fallback Sources:** Wikipedia for reliable information when search fails

The collected data is processed and filtered before being passed to the AI models for analysis and summarization.

1.7 Chapter Scheme

The report is organized into multiple chapters, each focusing on a specific aspect of the project:

- **Chapter 1:** Introduction, objectives, and overview of the system
- **Chapter 2:** System architecture, design, and workflow
- **Chapter 3:** Implementation details of frontend, backend, and AI agents
- **Chapter 4:** DevOps practices including Docker, CI/CD, and deployment
- **Chapter 5:** Database and storage mechanisms
- **Chapter 6:** Results and discussion of system performance
- **Chapter 7:** Summary and conclusions
- **Chapter 8:** Future scope and enhancements

Chapter 2: System Design & Investigation

2.1 System Architecture

The Energy Research App is designed using a modular, scalable, and microservices-based architecture that integrates frontend interaction, API handling, and AI-driven processing.

The system is divided into three major layers:

1. Presentation Layer (Frontend)

- Responsible for: Accepting user queries, Displaying research progress, Rendering structured reports
- Communicates with backend via HTTP APIs

2. Application Layer (API Server)

- Acts as a middleware/proxy layer
- Handles: Routing (/api/*), Request forwarding, CORS and security

3. Intelligence Layer (AI Backend)

- Implements Agentic AI architecture
- Consists of:
    - **Research Agent** → performs web search
    - **Scraper Tool** → extracts content
    - **Analysis Agent** → processes data
    - **Summary Agent** → generates final report using LLM

4. Data Layer

- FAISS → vector storage for semantic search
- JSON storage → stores generated reports
- Embeddings → Sentence Transformers

5. Deployment Layer

- Dockerized services
- Hosted on AWS EC2
- CI/CD via GitHub Actions

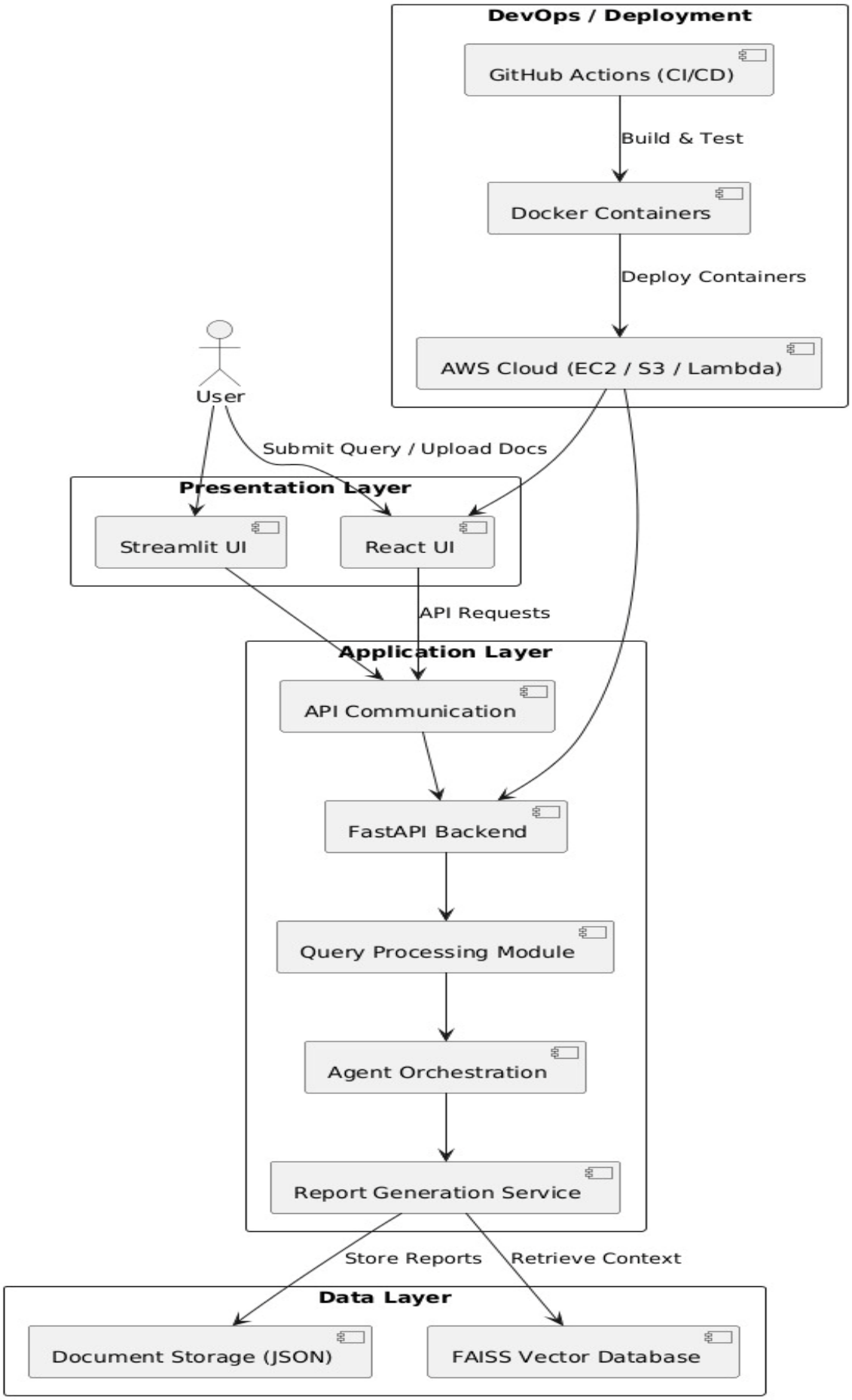

Fig. 2.1: Overall System Architecture

2.2 Technology Stack

The system integrates modern technologies across frontend, backend, AI, and DevOps layers.

Frontend Technologies - React (Vite), Tailwind CSS, TypeScript

Backend Technologies - Node.js (Express) – API Gateway, FastAPI – AI backend

AI & Machine Learning - HuggingFace Transformers (Flan-T5), Sentence Transformers (Embeddings), FAISS (Vector Database)

Data Processing - BeautifulSoup – Web scraping, Requests – HTTP handling, DuckDuckGo Search API

DevOps & Deployment - Docker – Containerization, Docker Compose – Service orchestration, GitHub Actions – CI/CD pipeline, AWS EC2 – Cloud deployment

2.3 System Workflow

The system follows a **multi-stage intelligent workflow** that automates the research process from query to report generation.

Step 1: User Input

- User submits a research query via frontend

Step 2: API Request Handling

- Request is sent to API server (/api/research)
- API server forwards request to Python backend

Step 3: Research Agent Execution

- Performs web search using DuckDuckGo
- Retrieves relevant URLs

Step 4: Content Scraping

- Scraper extracts data from web pages
- Cleans and structures content

Step 5: Analysis Phase

- Analysis agent filters noise
- Extracts key insights and topics

Step 6: Report Generation

- Summary agent uses LLM (Flan-T5)
- Generates structured research report

Step 7: Storage & Indexing

- Report stored in JSON format

- Embeddings generated and stored in FAISS

Step 8: Response Delivery

- Backend returns job ID
- Frontend fetches results using polling
- Final report displayed to user

Fig. 2.3: System Workflow Diagram

Chapter 3: Implementation

3.1 Frontend Implementation

The frontend of the Energy Research App is developed using React with Vite, providing a fast and responsive user interface.

3.1.1 API Integration

The frontend communicates with backend services using a generated API client and a custom fetch wrapper.

- Base URL is configured using: setBaseUrl("http://<EC2-IP>:8080");
- All API calls are made using:
    - React Query (for caching & async handling)
    - Custom fetch wrapper for request control

3.1.2 Key Features Implemented

- Query input interface
- Research history display
- Semantic search interface

3.2 Backend Implementation

The backend consists of two layers:

1. API Server (Node.js)
2. AI Backend (FastAPI)

3.2.1 API Server (Node.js - Express)

The API server acts as a gateway between frontend and AI backend.

Responsibilities:

- Route /api/* requests
- Forward to Python backend
- Handle CORS issues
- Maintain separation of concerns

Example: /api/research → forwarded to → Python /research

3.2.2 Python Backend (FastAPI)

The Python backend handles all AI processing and business logic.

3.2.3 API Endpoints Implemented

Method	Endpoint	Description
GET	/healthz	Health check
POST	/research	Start research
GET	/research/{id}	Get result
GET	/history	Fetch reports
POST	/search	Semantic search

3.3 AI Agent Implementation

The core intelligence of the system is based on a multi-agent architecture, where each agent performs a specific task in the research pipeline.

3.3.1 Research Agent

- Performs web search using DuckDuckGo
- Collects top URLs
- Initiates scraping process

3.3.2 Analysis Agent

- Filters irrelevant data
- Extracts key insights
- Identifies main topics

3.3.3 Summary Agent

- Uses HuggingFace Flan-T5 model
- Generates structured reports:
    - Introduction
    - Key insights
    - Conclusion

3.3.6 Agent Pipeline Flow

The agents operate sequentially:

1. Research Agent → gathers data
2. Scraper → extracts content
3. Analysis Agent → processes data
4. Summary Agent → generates report

Chapter 4: DevOps & Deployment

4.1 Dockerization

To ensure portability, consistency, and scalability, the entire application is containerized using **Docker**. Each major service runs inside its own isolated container.

4.1.1 Multi-Container Architecture

The system is divided into three Docker services:

- **Frontend Container**
  - Served via Nginx
  - Runs on port 80
- **API Server Container**
  - Handles routing and proxy logic
  - Runs on port 8080
- **Python Backend Container**
  - Executes multi-agent pipeline
  - Runs on port 8000

4.1.2 Docker Compose Orchestration

All services are managed using **Docker Compose**, enabling:

- Service networking
- Environment isolation
- Easy startup/shutdown

Example: services: frontend: api-server: python-backend:

4.2 CI/CD Pipeline (GitHub Actions)

The project implements a **Continuous Integration and Continuous Deployment (CI/CD)** pipeline using **GitHub Actions**.

4.2.1 Pipeline Overview: The pipeline automates:

- Code integration
- Docker image building
- Image publishing
- Remote deployment

4.2.2 Workflow Steps

1. Developer pushes code to GitHub (main branch)

2. GitHub Actions is triggered
3. Docker images are built:
    - energy-frontend
    - energy-api-server
    - energy-python-backend
4. Images are pushed to Docker Hub
5. EC2 instance is accessed via SSH
6. Latest images are pulled
7. Containers are restarted using Docker Compose

4.3 AWS EC2 Deployment

The application is deployed on a **cloud-based virtual server using AWS EC2**, enabling remote access and real-time usage.

4.3.1 EC2 Instance Setup

- OS: Ubuntu Linux
- Docker & Docker Compose installed

4.3.2 Deployment Process

1. EC2 instance is launched
2. SSH access configured using key pair
3. Docker environment installed
4. Application deployed via CI/CD pipeline
5. Services started using Docker Compose

4.3.3 Application Access

- Frontend accessed via: http://<EC2-PUBLIC-IP>
- API server: http://<EC2-PUBLIC-IP>:8080

4.4 Summary of DevOps Implementation

The integration of Docker, CI/CD, and AWS enables:

- Automated deployment pipeline
- Scalable architecture
- Reduced manual errors
- Faster development-to-production cycle

Chapter 5: Database & Storage

5.1 FAISS Vector Database

The system uses FAISS as a vector database to enable semantic search over stored research reports.Traditional databases rely on keyword matching, which is limited. FAISS allows:

- Semantic similarity search
- Faster retrieval of relevant information
- Efficient handling of high-dimensional embeddings

5.1.1 Working Principle

1. Text data is converted into vector embeddings using Sentence Transformers
2. These embeddings are stored in FAISS index
3. When a query is received:
    - It is converted into a vector
    - FAISS finds closest matching vectors using similarity search

5.2 JSON Storage System

In addition to FAISS, the system uses a JSON-based storage mechanism for storing complete research reports.

- Store structured research outputs
- Maintain history of generated reports
- Provide quick retrieval without recomputation

5.2.2 Storage Structure

Each report is stored as a JSON file containing: Report ID, Query, Generated summary, Sources, Timestamp

Example structure:

```
{
  "id": "report_id",
  "query": "What is renewable energy?",
  "summary": "...",
  "sources": ["url1", "url2"],
  "timestamp": "2026-04-21"
}
```

Chapter 6: Results and Discussions

6.1 System Output

The Energy Research App successfully generates structured research reports from user queries using its multi-agent pipeline.

6.1.1 Output Generation

When a user submits a query:

- The system processes it through multiple agents
- Extracts relevant information from web sources
- Generates a structured report containing:
    - Introduction
    - Key insights
    - Summary
    - References

6.1.2 Sample Output Description

For example, for the query:

"What is energy?"

The system produces:

- Definition and explanation
- Types of energy
- Real-world applications
- References from multiple sources

6.1.3 Frontend Output Display

The frontend displays:

- Research progress (live updates)
- Final structured report
- History of previous searches
- Semantic search results

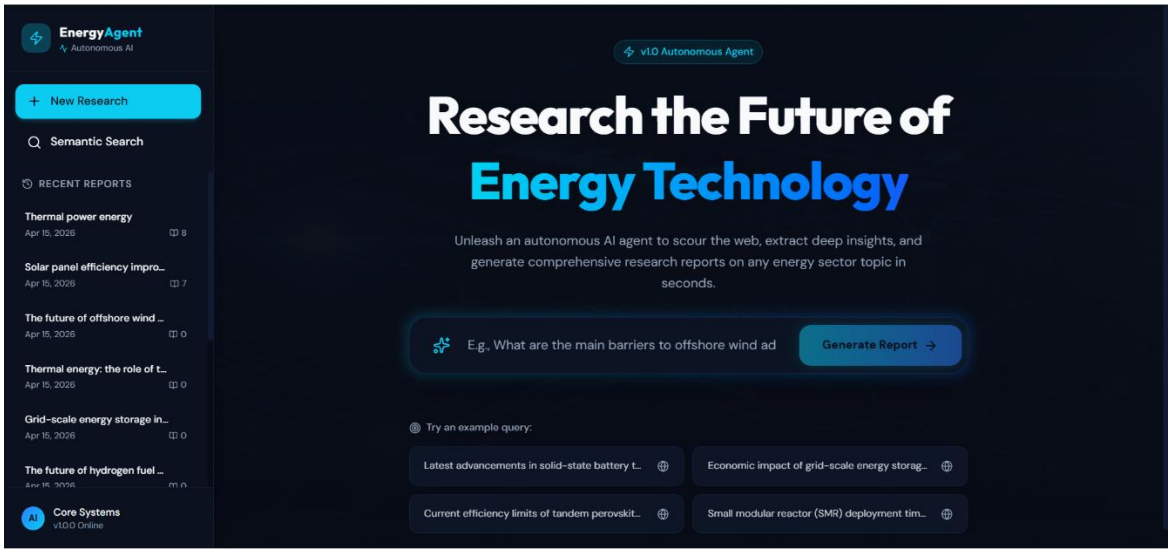

Fig. 6.1: User Interface – Query Input Page

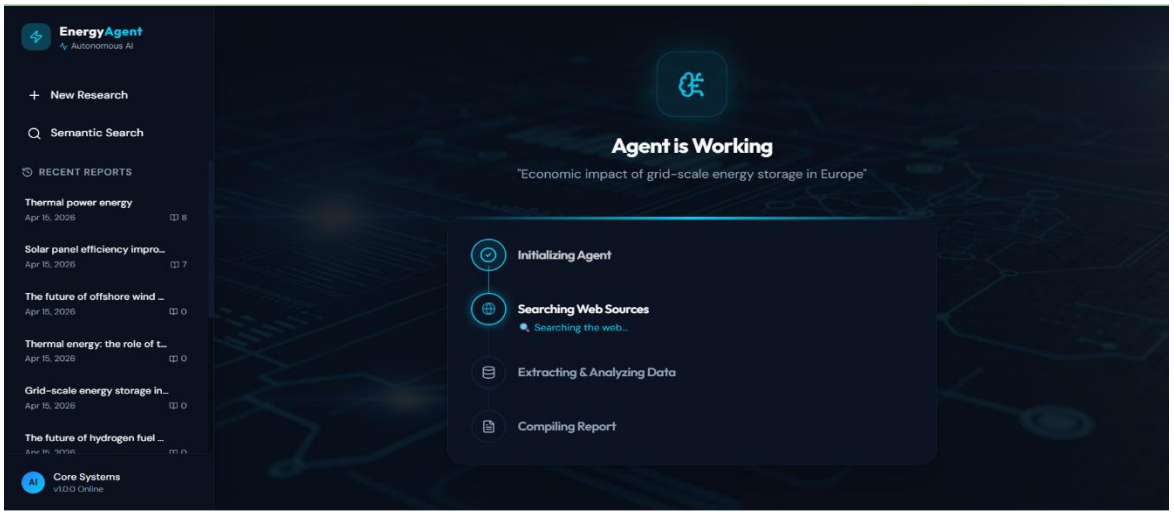

Fig 6.2: User Interface – Query Parsing Page

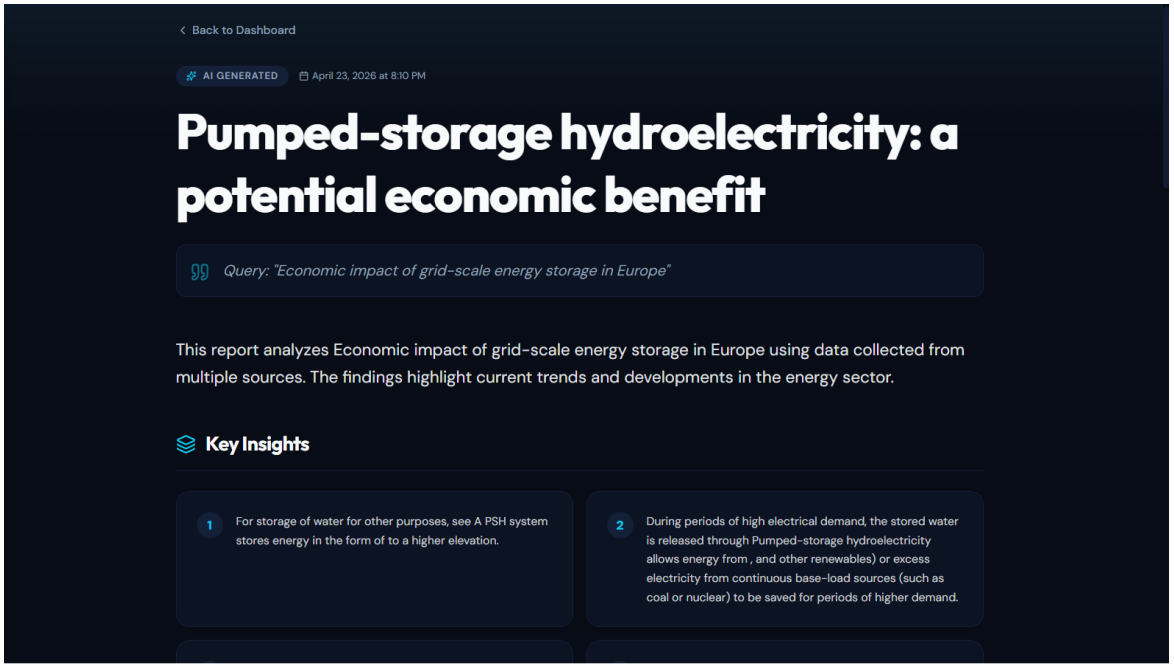

Fig 6.3: Generated Research Report Output Page (a)

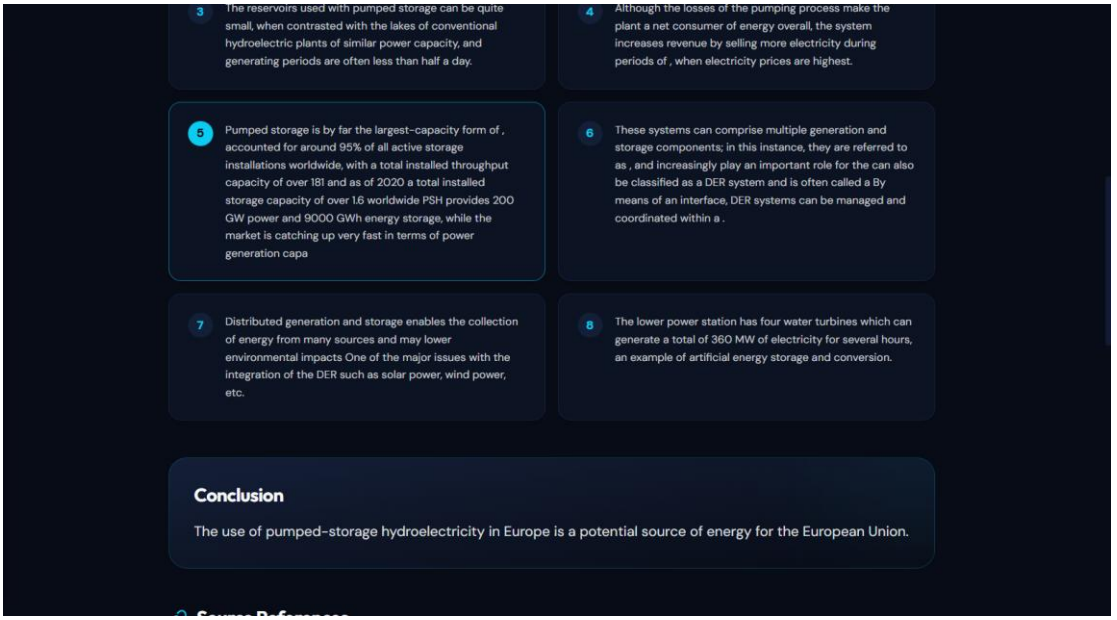

Fig 6.4: Generated Research Report Output Page (b)

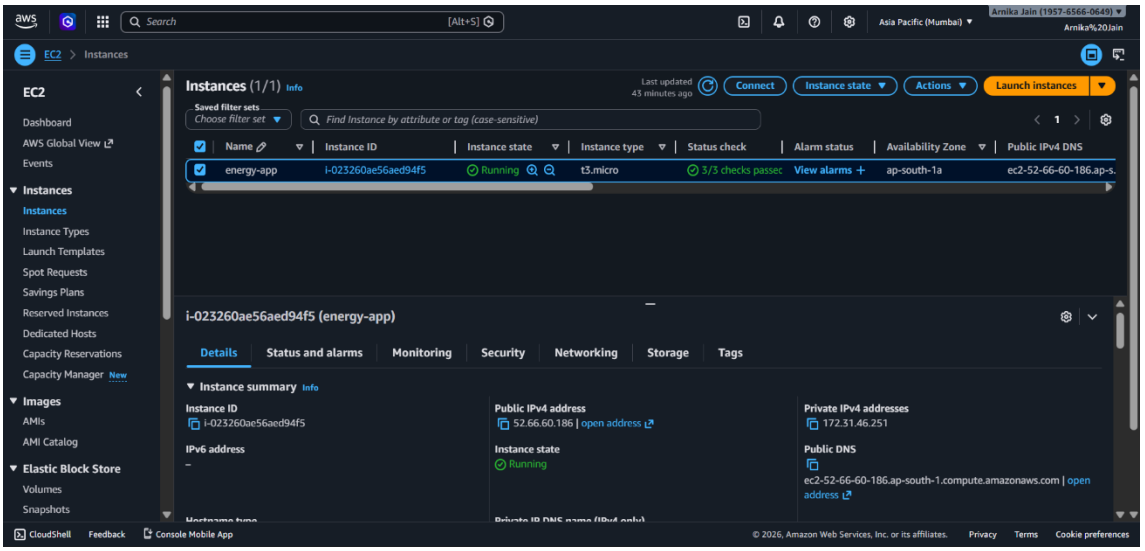

Fig 6.5: AWS EC2 Instance

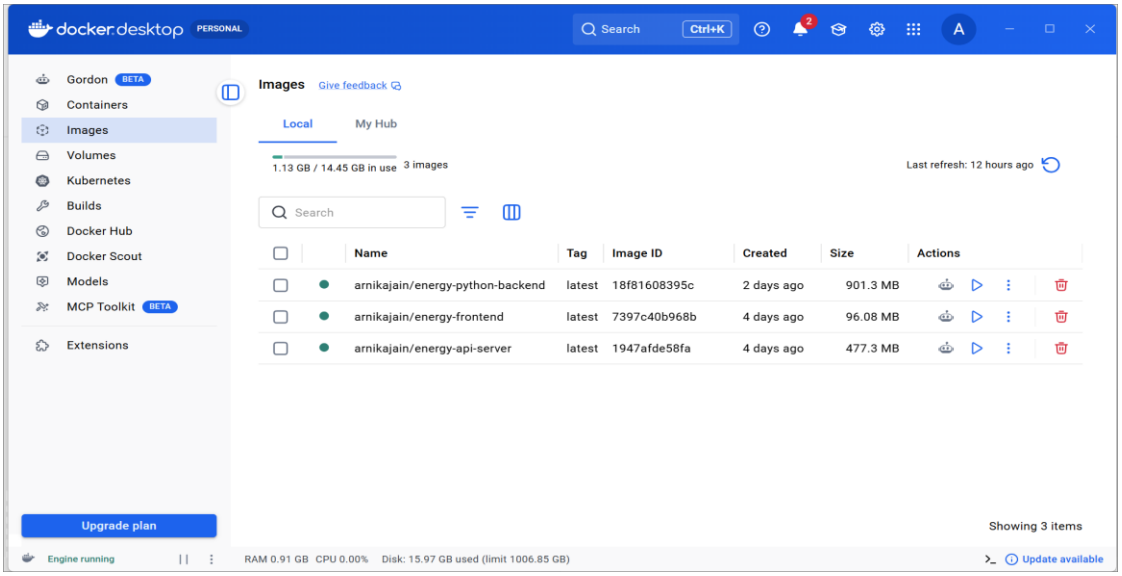

Fig 6.6: Docker Desktop

Chapter 7: Summary and Conclusion

7.1 Summary

The application follows a multi-agent architecture, where:

- The Research Agent collects data
- The Analysis Agent processes content
- The Summary Agent generates structured reports

Challenges Faced

During development and deployment, several technical challenges were encountered.

API Routing Issues:

Problem: 404 errors when frontend tried to access backend

Solution: Fixed API proxy configuration, Corrected base URL in frontend

Deployment Issues on Cloud:

Problem: Application worked locally but not on EC2

Solution: Updated Docker configuration, Corrected exposed ports and URLs

Disk Space Issues:

Problem: EC2 storage getting full due to multiple images

Solution: Implemented cleanup command: docker image prune -af

7.2 Conclusion

The Energy Research App successfully demonstrates how Artificial Intelligence, Agentic Systems, and DevOps practices can be integrated into a unified application.

Key conclusions drawn from the project:

- The system is capable of generating accurate and structured research outputs
- The multi-agent approach improves modularity and efficiency
- Semantic search using FAISS enhances data retrieval capabilities
- Dockerization and CI/CD enable smooth and automated deployment
- The system is scalable and production-ready

Despite minor challenges during development, such as API routing issues and deployment errors, the final system operates reliably and efficiently.

Overall, the project showcases a modern AI-driven application architecture combining Machine Learning, Web Technologies, Cloud Deployment, Automation (DevOps)

Chapter 8: Future Scope

The current system provides a strong foundation for automated research; however, several enhancements can be implemented to improve functionality and scalability.

1 Advanced AI Models

- Integration of more powerful LLMs (e.g., GPT-based or larger transformer models)
- Improved summarization quality and contextual understanding

2 Real-Time Data Integration

- Use of live APIs for real-time data sources
- Integration with academic databases (e.g., research papers)

3 User Authentication System

- Login and user-specific dashboards
- Personalized research history

4 Scalability Improvements

- Deployment using Kubernetes for container orchestration
- Load balancing for handling multiple users

5 Enhanced Semantic Search

- Hybrid search (keyword + vector search)
- Improved ranking algorithms

6 UI/UX Improvements

- Interactive dashboards
- Visualization of research data (charts, graphs)

7 Mobile Application

- Development of Android/iOS version
- Cross-platform support

8 Integration with External Tools

- Integration with citation tools
- Export reports as PDF or DOCX

9 Performance Optimization

- Caching mechanisms
- Faster scraping techniques
- Model optimization

10 Security Enhancements

- API authentication (JWT)
- Secure data handling

Appendix

Appendix A: Sample API Endpoints

Method	Endpoint	Description
GET	/healthz	Health check
POST	/research	Start research
GET	/research/{id}	Get result
GET	/history	Fetch reports
POST	/search	Semantic search

Appendix B: Sample Docker Commands

```
docker compose up -d
docker compose down
docker image prune -af
```

Appendix C: Project Structure

```
artifacts/
 ├── python-backend/
 ├── api-server/
 └── energy-researcher/
```

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